

Name: _____ Student ID: _____

COMPLEX ANALYSIS – FINAL

This exam has 2 pages. Before starting the exam, please check that your copy contains all 2 of them and obtain a new copy of the exam immediately if it does not.

No books, notes, or calculators are allowed. **100 Points, 180 Minutes.**

1. (20 points) True or False. (Just mark **True** or **False** for the following statements.)

(1) Given $\Omega \subset \mathbb{C}$ a bounded domain, $f \in C^0(\partial\Omega)$, the Dirichlet problem with respect to Ω , f always has a solution.

(2) Any holomorphic function $f : \mathbb{C} \rightarrow \mathbb{C} \setminus [0, \infty)$ must be constant.

(3) Given $\Omega \subset \mathbb{C}$ and a (real) harmonic function u on Ω , there always exists a holomorphic function f such that $u = \operatorname{Re}(f)$.

(4) The domains $\mathbb{D}^*, \mathbb{C}^*$ are conformal.

(5) There exists a non-constant function f which is constant on the arc $I = \{e^{i\theta} | 0 < \theta < \pi/4\}$ and is continuous on $\overline{\mathbb{D}}$ and holomorphic in $\mathbb{D}$.

(6) There exists a meromorphic function on $\mathbb{C}$ such that its zeros are exactly $\{2k | k \in \mathbb{Z}\}$ and its poles are exactly $\{2k + 1 | k \in \mathbb{Z}\}$.

(7) If f is holomorphic in the first quadrant $\{z = x + iy | x > 0, y > 0\}$ and $f(\frac{1+i}{n}) = 0$ for every positive integer n , then f is constant.

(8) A connected open set $\Omega \subset \mathbb{C}$ is simply connected if $\log f$ is well defined for any non-vanishing holomorphic function f on Ω .

(9) The domains $\mathbb{C} \setminus \{1, 2\}$, $\mathbb{C} \setminus \{100, 2i\}$ are conformal.

(10) Let f_k be a sequence of continuous subharmonic functions on Ω . Assume f_k converges to f uniformly on every compact subsets, then f is also subharmonic.

2. (10 points) Let $u : \Omega \rightarrow \mathbb{R}$ be a continuous function. Prove that u is harmonic (i.e. $\Delta u = 0$) if and only if u satisfies the mean value property, that is, for any closed disc $\overline{D(p; r)} \subset \Omega$ we have

$$u(p) = \frac{1}{2\pi} \int_0^{2\pi} u(p + re^{i\theta}) d\theta.$$

3. (10 points) Explain how the Gamma function $\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt$, which is initially defined for $s > 0$, extends to a meromorphic function on $\mathbb{C}$. Find all of its poles and its residues at these poles.

4. (5 points) Let $\Omega \subset \mathbb{C}$ be a proper simply connected domain. Explain how to construct an injective holomorphic map $F : \Omega \rightarrow \mathbb{C}$ such that $F(\Omega)$ is bounded. (You cannot directly apply Riemann mapping theorem here.)

5. (10 points) Give an explicit construction of a conformal mapping which maps $\{z \in \mathbb{C} | |z - 1| < 2, |z + 1| < 2\}$ to the unit disc $\mathbb{D}$.

6. (10 points) Prove the identity:

$$\frac{1}{\pi} \sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 1} = \sum_{n=-\infty}^{\infty} e^{-2\pi|n|}.$$

7. In this problem, study the complex geometry of the unit disc and its application.

(1) (10 points) Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be a holomorphic map, show that for any $z \in \mathbb{D}$,

$$\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2}.$$

(Hint: reduce it to Schwarz lemma.)

Endow the unit disc $\mathbb{D}$ with the Poincaré metric $ds^2 = \frac{|dz|^2}{(1-|z|^2)^2}$, in which sense the Poincaré length of a smooth curve $\gamma : [a, b] \rightarrow \mathbb{D}$ is defined by

$$L(\gamma) = \int_a^b \frac{|\gamma'(t)|}{1 - |\gamma(t)|^2} dt,$$

where $\gamma'(t) = \frac{d}{dt}\gamma(t)$.

(2) (2 points) Given $0 < R < 1$, find the Poincaré length of the curve $\gamma : [0, R] \rightarrow \mathbb{D}$, $\gamma(t) = t$.

Given two points $p, q \in \mathbb{D}$, the Poincaré distance $d(p, q)$ is defined by

$$d(p, q) = \inf_{\gamma} L(\gamma),$$

where the infimum is taken over all smooth curves $\gamma : [a, b] \rightarrow \mathbb{D}$ with $\gamma(a) = p, \gamma(b) = q$. A curve connecting p, q is called a geodesic if its Poincaré length equals the distance $d(p, q)$.

(3) (3 points) Find $d(0, R)$, where $0 < R < 1$.

(4) (2 points) Prove that $d(f(p), f(q)) \leq d(p, q)$ holds for any holomorphic map $f : \mathbb{D} \rightarrow \mathbb{D}$.

(5) (5 points) Given two points $p, q \in \mathbb{D}$, find the formula of $d(p, q)$.

(6) (3 points) Given two points $p, q \in \mathbb{D}$, show that there always exists a geodesic connecting them.

Let $\{p_k\}_{k=1}^{\infty} \subset \mathbb{D}$ be a sequence of points. It is called a Cauchy sequence with respect to Poincaré distance if for any $\varepsilon > 0$, there exists N such that $d(p_m, p_n) < \varepsilon$ whenever $m, n \geq N$.

(7) (5 points) Assume that the sequence $\{p_k\}_{k=1}^{\infty} \subset \mathbb{D}$ is Cauchy with respect to Poincaré distance, prove that there is a unique $q \in \mathbb{D}$ such that

$$\lim_{k \rightarrow \infty} d(p_k, q) = 0.$$

(In particular, $\mathbb{D}$ is called a complete metric space with respect to Poincaré distance.)

(8) (5 points) Assume that $f : \mathbb{D} \rightarrow \mathbb{D}$ is a holomorphic map such that the closure of $f(\mathbb{D})$ is compact in $\mathbb{D}$, show that there is a unique point $q \in \mathbb{D}$ such that $f(q) = q$.

(Hint: you can use the following fact: a self-contraction of a complete metric space has a unique fixed point. Thus in the above setting you only need to show: there is $0 < \delta < 1$ such that $d(f(p), f(q)) \leq \delta d(p, q)$. For a fixed $z_0 \in \mathbb{D}$, consider $g(z) = f(z) + \varepsilon(f(z) - f(z_0))$ for some suitable $\varepsilon > 0$ such that g maps $\mathbb{D}$ to $\mathbb{D}$.)

Conventions and notions:

$$\mathbb{C}^* = \{|z| > 0\}, \mathbb{D} = \{|z| < 1\}, \mathbb{D}^* = \{0 < |z| < 1\}.$$